Pooya Barazandeh

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Summary

Computer Science graduate ranked 1st out of 72 students at Amirkabir University of Technology with a GPA of 19.55/20, specializing in AI/ML, federated learning, LLM/RAG systems, and computer vision. Authored a submitted paper on communication-efficient vertical federated learning and completed Oxford Machine Learning School's MLx Representation Learning & Generative AI 2026 programme. Teaching Assistant for 10+ undergraduate and graduate courses, including Head TA for Advanced AI, with industry experience leading production AI and developer-tooling projects for EdTech platforms serving 40+ million users. TOEFL iBT: 5/6 (100/120 equivalent).

Education

Amirkabir University of Technology (Tehran Polytechnic) Tehran, Iran
B.Sc. in Computer Science Sep 2022 - Jul 2026
- GPA: 19.55/20 (Equivalent to 4.0/4.0) - Rank: 1st out of 72 students

Allameh Helli 8 High School (NODET/SAMPAD) Tehran, Iran
Diploma in Experimental Sciences Sep 2017 - Jun 2021
- GPA: 19.23/20 (Top-ranked talented student school)

Research Experience

Submitted Manuscript: Gradient-Noise-Scale-Aware Adaptive Sampling for Communication-Efficient Vertical Federated Logistic Regression Feb 2026 - Present
Amirkabir University
- Co-authored a submitted paper proposing a gradient-noise-scale-aware adaptive sampling controller for communication-efficient vertical federated logistic regression.
- Implemented and analyzed 950 runs across 10 datasets and 5 seeds; the reference setting reduced sample-party communication by 90.89% at macro ROC-AUC 0.729.

Bachelor's Thesis: Deep Learning for Knowledge Tracing Sep 2025 - Jun 2026
Supervised Research Project Amirkabir University
- Designed and evaluated transformer- and graph-based Knowledge Tracing models (DKT, AKT, GKT) on ASSISTments and EdNet for personalized learning prediction.
- Investigated attention mechanisms for interpretable modeling of student knowledge states.

Research Assistant (Part-time) Jan 2025 - Jan 2026
NORC, Amirkabir University of Technology Iran (Hybrid)
- Conducted computer vision research on the Urban Bridge Failure Detection System under the supervision of Prof. Mehdi Ghatee.
- Advanced the bridge-defect detection project in collaboration with Tehran Urban Planning and Research Center (TUPRC), using Python, PyTorch, OpenCV, and YOLO to identify structural defects in inspection images.

AI R&D Engineer - Applied Research Sep 2024 - Present
Rastanovin Raya (Ministry of Education Contractor, 200+ employees) Tehran, Iran
- Conducting applied research on production-grade AI/ML systems powering educational platforms with 40+ million users.
- Developing LLM-based automation pipelines reducing manual document processing time by 60%.
- Building intelligent assistant systems integrating RAG architectures with domain-specific knowledge bases for large-scale EdTech applications.

Teaching Experience

Teaching Assistant - 10+ Undergraduate & Graduate Courses Jan 2023 - Jul 2026
Amirkabir University of Technology Tehran, Iran

- Courses: Data Structures & Algorithms, Artificial Intelligence, Numerical Linear Algebra, Operating Systems, Foundations of Mathematics, Basic Programming, Logic, Principles of Computer Systems, Computer Simulation, Network Security, Introductory Probability & Statistics, Image Processing, and Data Mining.
- Head TA for Advanced Artificial Intelligence (M.Sc. course): Led TA team for 40+ graduate students.
- Designed 15+ programming assignments and projects; mentored 200+ students across multiple semesters.

Teaching Assistant - Operating Systems Sep 2025 - Jul 2026
University of Tehran Tehran, Iran

- Provided hands-on lab support for process management, memory allocation, and file systems projects.
- Mentored 40+ students on systems programming assignments in C.

Work Experience

AI Developer & Team Lead (Part-time) Sep 2024 - Present
Rastanovin Raya - Ministry of Education Contractor (200+ employees) Tehran, Iran

- Developing AI solutions for Iran's largest EdTech platform serving 40+ million users (students, teachers, administrators) through Ministry of Education contracts.
- Led a team of 2 developers to build Vibe Coding features for the company's Cloud Development Environment (CDE) - an internal platform handling CI/CD, code hosting, and full-stack development workflows.
- The CDE New Style demo version is currently used by 30+ developers building complete applications (frontend, backend, database) from scratch to production deployment.
- Built LLM-based automation pipelines for intelligent code assistance, document processing, and developer productivity tools.
- Designed ML model pipelines achieving 92% accuracy on structured document parsing at scale.

Frontend Developer Intern Jul 2023 - Sep 2023
Rastanovin Raya Tehran, Iran

- Developed responsive web interfaces using React and JavaScript for internal enterprise tools.
- Implemented RESTful API integrations and improved UI/UX based on user feedback.

Selected Projects

Cloud Development Environment (CDE) - Vibe Coding Features | Python, JS, LLM, Docker Oct 2024 - Present

- Led development of AI-powered coding features for enterprise cloud IDE serving 30+ active developers.
- Implemented context-aware code generation, intelligent autocomplete, and automated code review using LLMs.
- Integrated with existing CI/CD pipelines enabling end-to-end development workflows (frontend, backend, database).

LLM Code Assistance Tool | Python, JavaScript, LangChain, Git Oct 2024 - Present

- Developed a Cursor-like AI coding assistant with context-aware prompt generation and automatic code merging.
- Implemented intelligent diff-based merge algorithm achieving 95%+ successful code integrations.

RAG System | Python, Docker, LangChain, React Mar 2025

- Built production-ready Retrieval-Augmented Generation pipeline indexing 10,000+ documents with vector embeddings.
- Integrated fine-tuned LLMs for domain-specific Q&A; Dockerized for scalable cloud deployment.

Exam Paper Segmentation & Blank Detection | Python, OpenCV Aug 2024

- Designed computer vision system for Ministry of Education automating exam paper processing.
- Achieved 96% accuracy in question boundary detection and blank answer identification across 5,000+ pages.

Urban Bridge Failure Detection System | Python, PyTorch, OpenCV, YOLO Feb 2024 - Present

- Industrial collaboration with Tehran Urban Planning and Research Center (TUPRC) on AI-driven infrastructure inspection.
- Developed deep learning pipeline detecting structural defects (cracks, corrosion, deformation) from bridge inspection images.

Sabad - E-Commerce Platform | React, Node.js, PostgreSQL Jul 2023 - Sep 2023

- Developed full-stack shopping platform with user authentication, product catalog, and order management.

Fraud & ADHD Detection Models | Python, scikit-learn, pandas Jun 2024

- Built classification pipelines for fraud detection and ADHD prediction tasks.

Selected Recent Projects

DemandCast | Python, LightGBM, Streamlit 2026
github.com/pooyabrz/demandcast

- Built retail-sales forecasting and a review dashboard with temporal backtesting and quantile intervals on 500 M5 item/store series over a 28-day horizon; compared median- and mean-oriented forecasts, uncertainty, and aggregate-sales bias.

LeakScope | Python, WNTR/EPANET, Streamlit 2026
github.com/pooyabrz/LeakScope

- Built a research prototype for ranking water-network inspection areas on simulated BattLeDIM data using hydraulic signatures, calibration, sensor planning, and dropout handling; evaluated sparse-sensor performance in a frozen controlled benchmark.

Nimbryva | Python, Scientific Simulation, Matplotlib, PPO 2026
github.com/pooyabrz/NIMBRYVA-ADAPTIVE-WATER-ENERGY-FACADE

- Built a rotating water/solar-facade concept demo with rigid geometry and a buffer-discharge state machine; preserved earlier thermal, water, PV, and trained-PPO research for replay. Revised-device physical performance remains unvalidated.

Awards & Certifications

- Certificate of Achievement - MLx Representation Learning & Generative AI 2026, Oxford Machine Learning School (OxML), Jul 2026 - completed an intensive 18-hour programme and associated assessment.
- Top 1% - National University Entrance Exam (Konkur) for B.Sc. Admission
- Qualified Candidate - Passed 1st Round of National Mathematics Olympiad (2 consecutive years)
- Top-Ranked Participant - Quera Fito Hackathon (National Programming Competition)
- Top Performer in Artificial Intelligence - 2nd National Technology Olympiad
- Top Performer in Frontend Development - 2nd National Technology Olympiad
- Algorithm Arena Certificate - Scrum/Agile Certification

University Activities

Scientific Association Member & Event Staff May 2024 - Jul 2026
Computer Science Department Amirkabir University

- Organized technical workshops on AI/ML topics with 100+ student attendees.
- Served as staff coordinator for university programming competitions and academic events.
- Mentored junior students on academic projects and career guidance.

Technical Skills

Programming Languages: Python, C/C++, JavaScript, TypeScript, SQL (PostgreSQL), R, LaTeX

AI/ML Areas: LLMs, RAG, Federated Learning, Knowledge Tracing, Computer Vision, Image Processing, Representation Learning, Generative AI

AI/ML Libraries: PyTorch, TensorFlow, scikit-learn, LangChain, Hugging Face Transformers, OpenCV

Frameworks & Tools: React, Node.js, Docker, Git, CI/CD, REST APIs

Data & Modeling: NumPy, pandas, Matplotlib, Jupyter, Data Mining, Probability & Statistics, Computer Simulation

Systems & Development: Linux/Unix, Operating Systems, Network Security, CI/CD, VS Code, PyCharm, CLion

Additional Project Tools: LightGBM, Streamlit, WNTR/EPANET, PPO

Languages

Persian: Native - English: Fluent (Academic & Professional Proficiency) - TOEFL iBT: 5/6 (100/120 equivalent)